

POINTERS

Pointer is a variable, which holds the address of another variable of same type.

Pointer is a memory location, which holds the address of another memory location.

Pointer is a derived data type.

Advantages:

1. Dynamic memory allocation.
2. Program performance is increased due to preventing memory wastage.
3. They are very much used in System programming.
4. They are very much used in dynamic linked list & Stacks [**data structures**].
5. **It allows to access local variable outside the function i.e. data sharing between functions.**
[**call by address/Reference**].
6. **To handle strings, arrays etc in functions we need pointers.**

7. To handle **data files** we are using pointers.
8. They directly works on variable address. Due to this search time is reduced and execution speed is increased.

Dis-advantage:

They are not secured.

Syntax:

datatype * variable;

- * indicates it is a pointer data type.
- * is called indirection operator.
- * is called dereferencing operator.
- *** is a re-direction operator.**
- * indicates value at that address.
- * indicates pointer value.

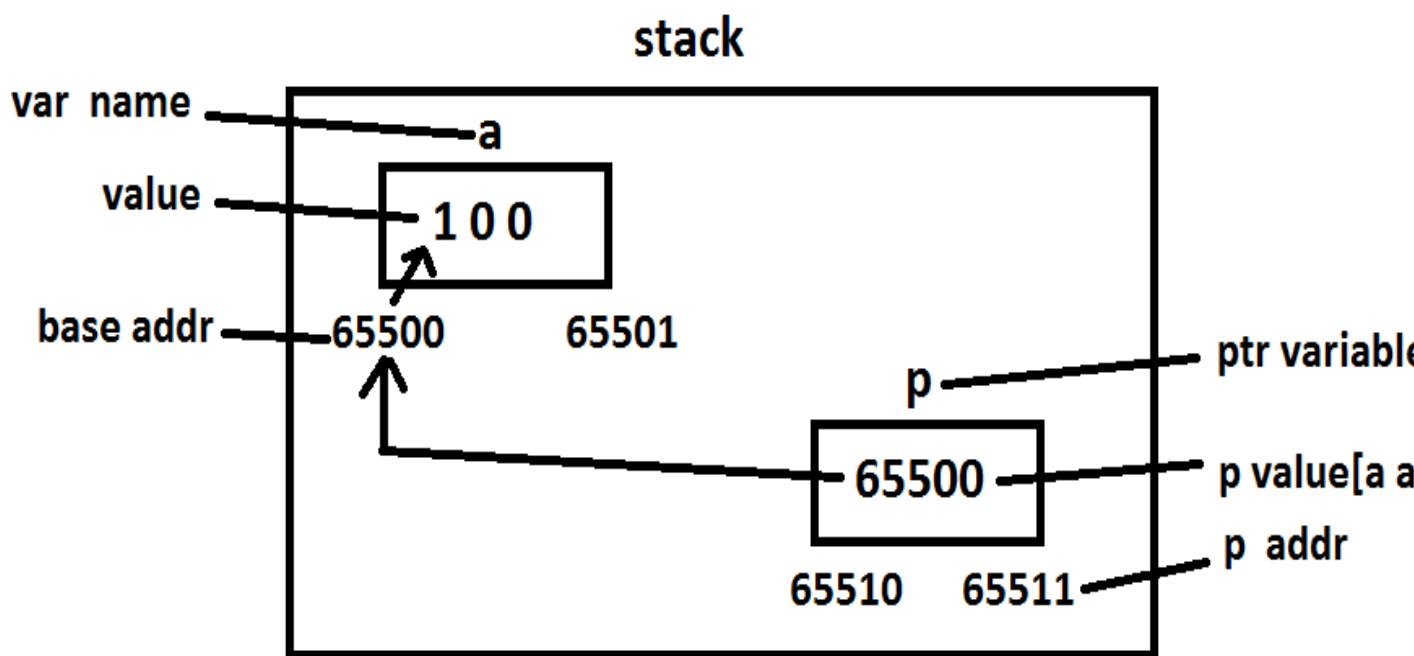
Eg:

```
int a=100, * p;
```

In the above example '**a**' is a general variable.

* indicates '**p**' is a pointer type variable and it is able to store the address of general variable '**a**' as follows.

```
p = &a;
```



In the above example, to pick the value of **a** through pointer variable **p**, we have to use the **printf()** as follows.

```
printf( "%d", *p );
```

→ **100**

Here ***p** means **value of p** or **value at that addr.** i.e. **65500**. But **65500** is the **addr of 'a'**. The **value in a address is 100**.

Or

Here **p** means **65500**. ***p** means **value at 65500**. i.e. **100**.

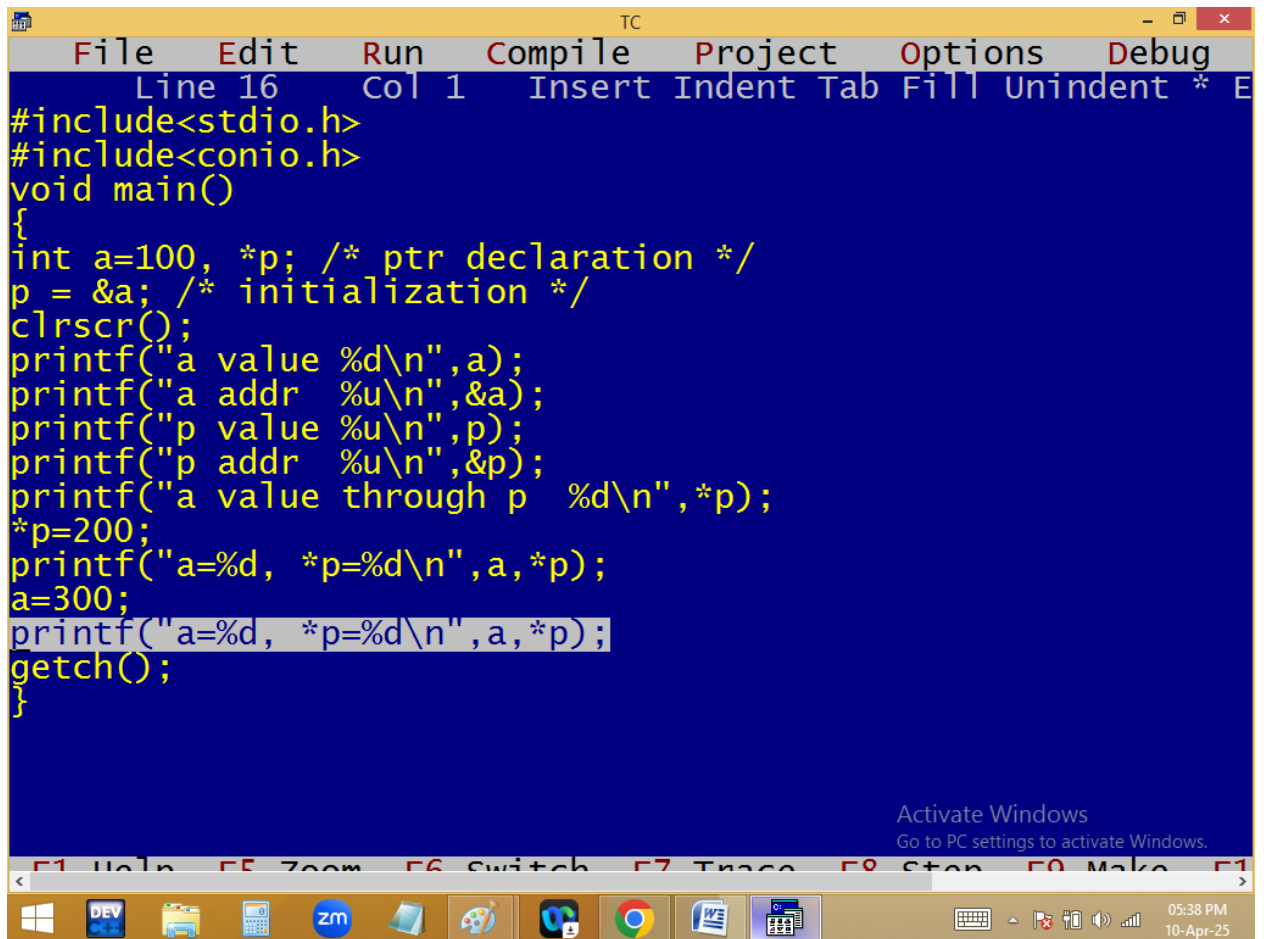
Due to this example any changes conducted in ***p** effects the value of **'a'**. Hence **p** is called **pointer to a**.

Eg: ***p=200;**

Now **a** value becomes **200**.

Eg:

Finding a variable value and address using a pointer:

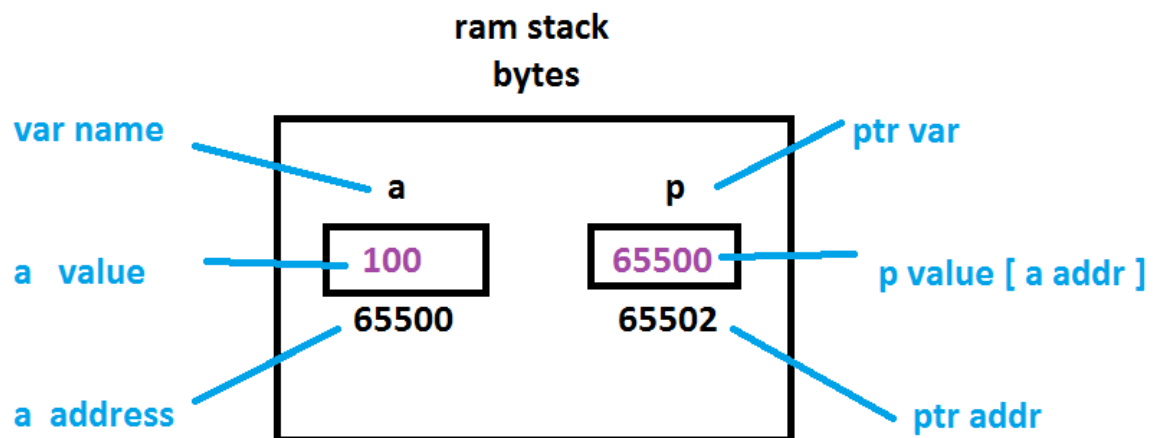


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 16", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a dark blue background with yellow text. It contains a C program that demonstrates pointer manipulation. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=100, *p; /* ptr declaration */
    p = &a; /* initialization */
    clrscr();
    printf("a value %d\n",a);
    printf("a addr %u\n",&a);
    printf("p value %u\n",p);
    printf("p addr %u\n",&p);
    printf("a value through p %d\n",*p);
    *p=200;
    printf("a=%d, *p=%d\n",a,*p);
    a=300;
    printf("a=%d, *p=%d\n",a,*p);
    getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, a folder, a calculator, a Zoom icon (zm), a paint application, a code editor, Google Chrome, a document, and a calculator. On the right side of the taskbar, the system clock shows "05:38 PM" and the date "10-Apr-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

```
TC
a value 100
a addr 65500
p value 65500
p addr 65502
a value through p 100
a=200, *p=200
a=300, *p=300
```

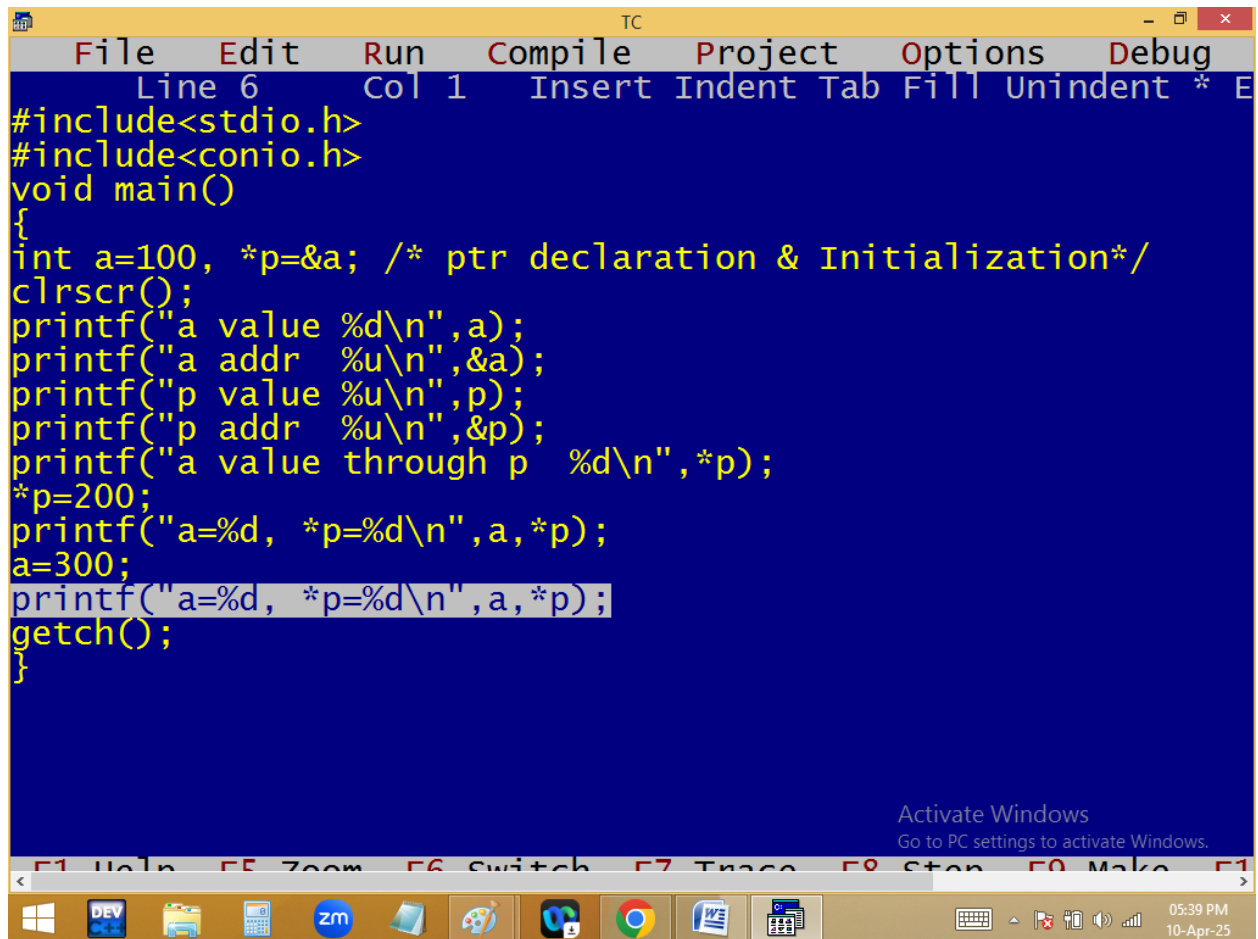


Least priority

`p("%d", *p); ==> 100`

`65500`

`*65500 ==> value at 65500 ==> 100`



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 1", and "Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, it declares an integer `a` with value 100 and a pointer `p` pointing to `a`. It then prints the value and address of `a`, and the value and address of `p`. It also prints the value of `a` through the pointer `p`. After changing `*p` to 200 and `a` to 300, it prints the new values. The code ends with `getch()` and a closing brace. The Windows taskbar is visible at the bottom with icons for Windows, DEV, File Explorer, Calculator, Zoom, and others. The system clock shows 05:39 PM on 10-Apr-25. An "Activate Windows" watermark is present in the bottom right corner of the IDE window.

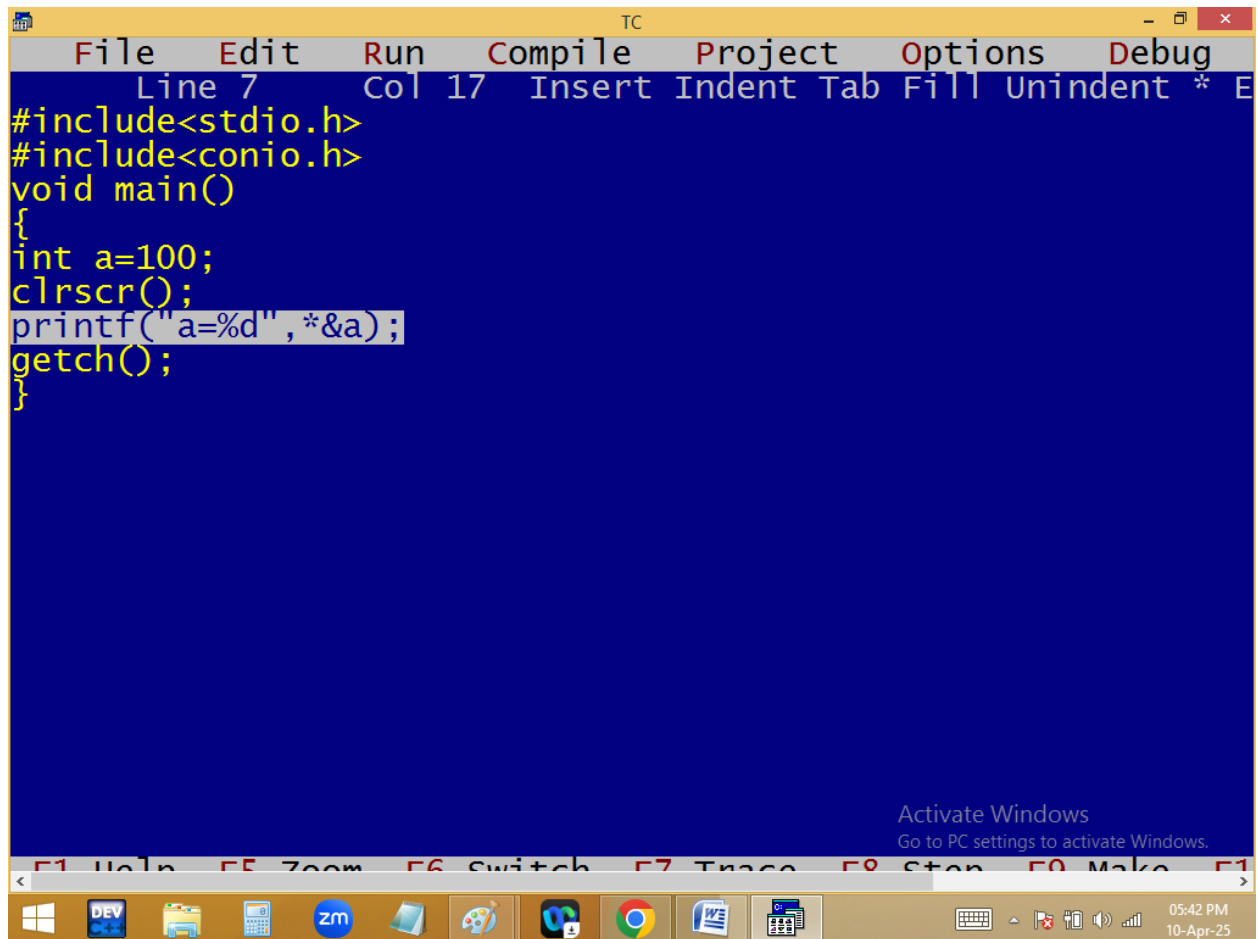
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a; /* ptr declaration & Initialization*/
clrscr();
printf("a value %d\n",a);
printf("a addr %u\n",&a);
printf("p value %u\n",p);
printf("p addr %u\n",&p);
printf("a value through p %d\n",*p);
*p=200;
printf("a=%d, *p=%d\n",a,*p);
a=300;
printf("a=%d, *p=%d\n",a,*p);
getch();
}
```

```
TC
a value 100
a addr 65500
p value 65500
p addr 65502
a value through p 100
a=200, *p=200
a=300, *p=300
```

Activate Windows
Go to PC settings to activate Windows.

05:39 PM
10-Apr-25

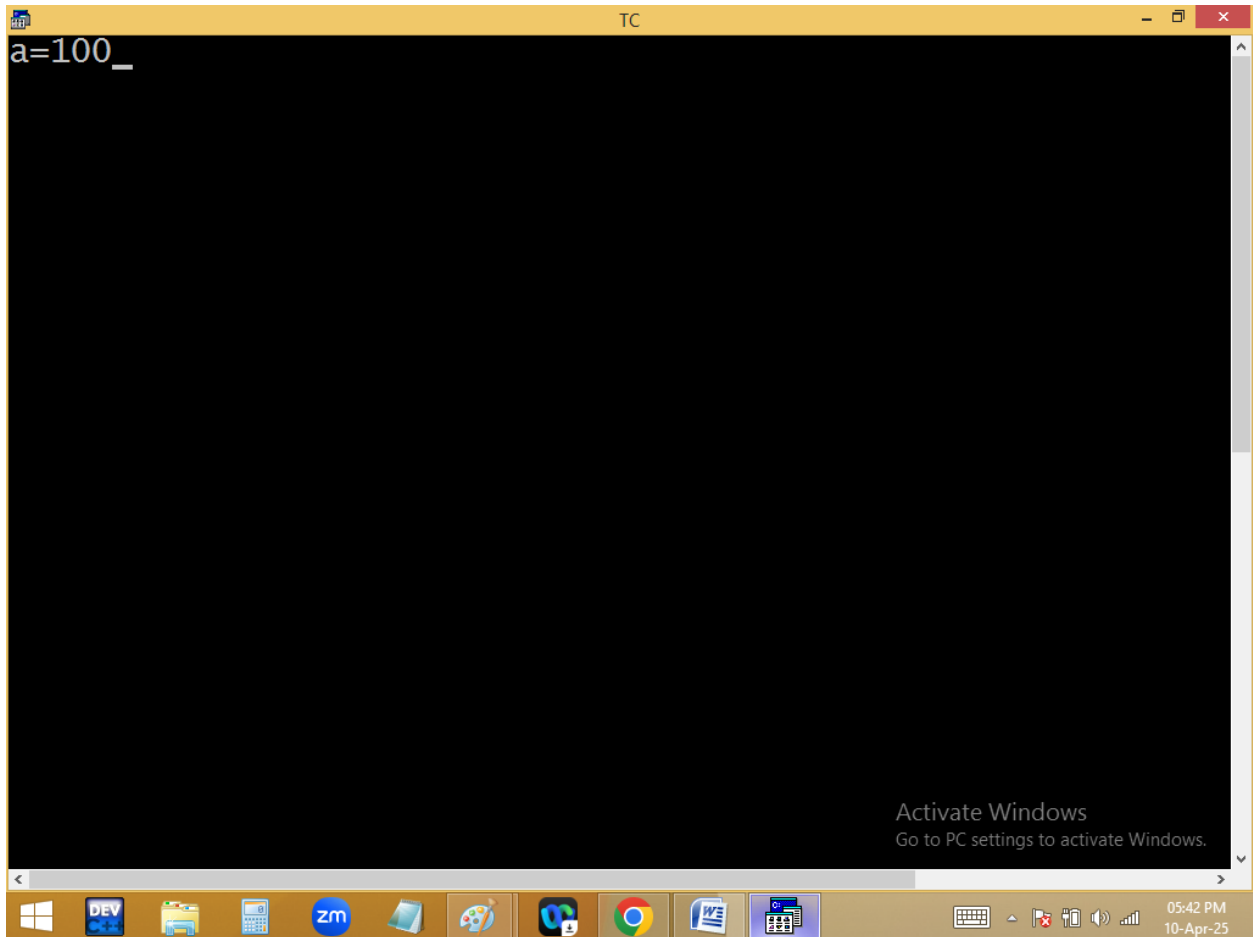
Printing a normal variable value using pointer mechanism?



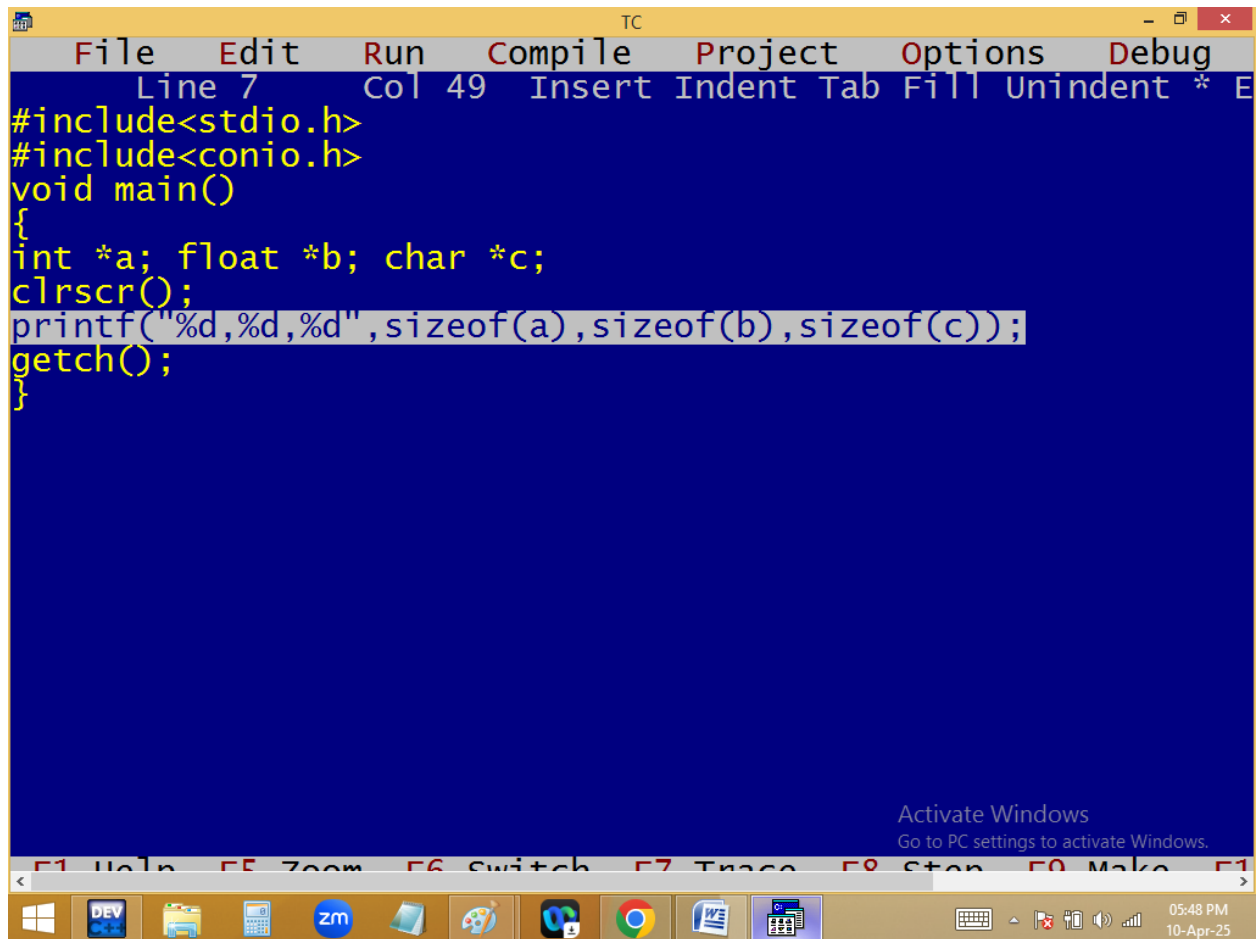
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 7" and "Col 17". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100;
clrscr();
printf("a=%d", *&a);
getch();
}
```

The text "printf("a=%d", *&a);" is highlighted. At the bottom of the IDE window, there is a text box that says "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the date and time as "05:42 PM 10-Apr-25".



Finding pointer size: Any type of pointer it stores only the base address and it is always unsigned int. due to this reason pointer takes 2/4/8 bytes in 16/32/64 bit compilers.

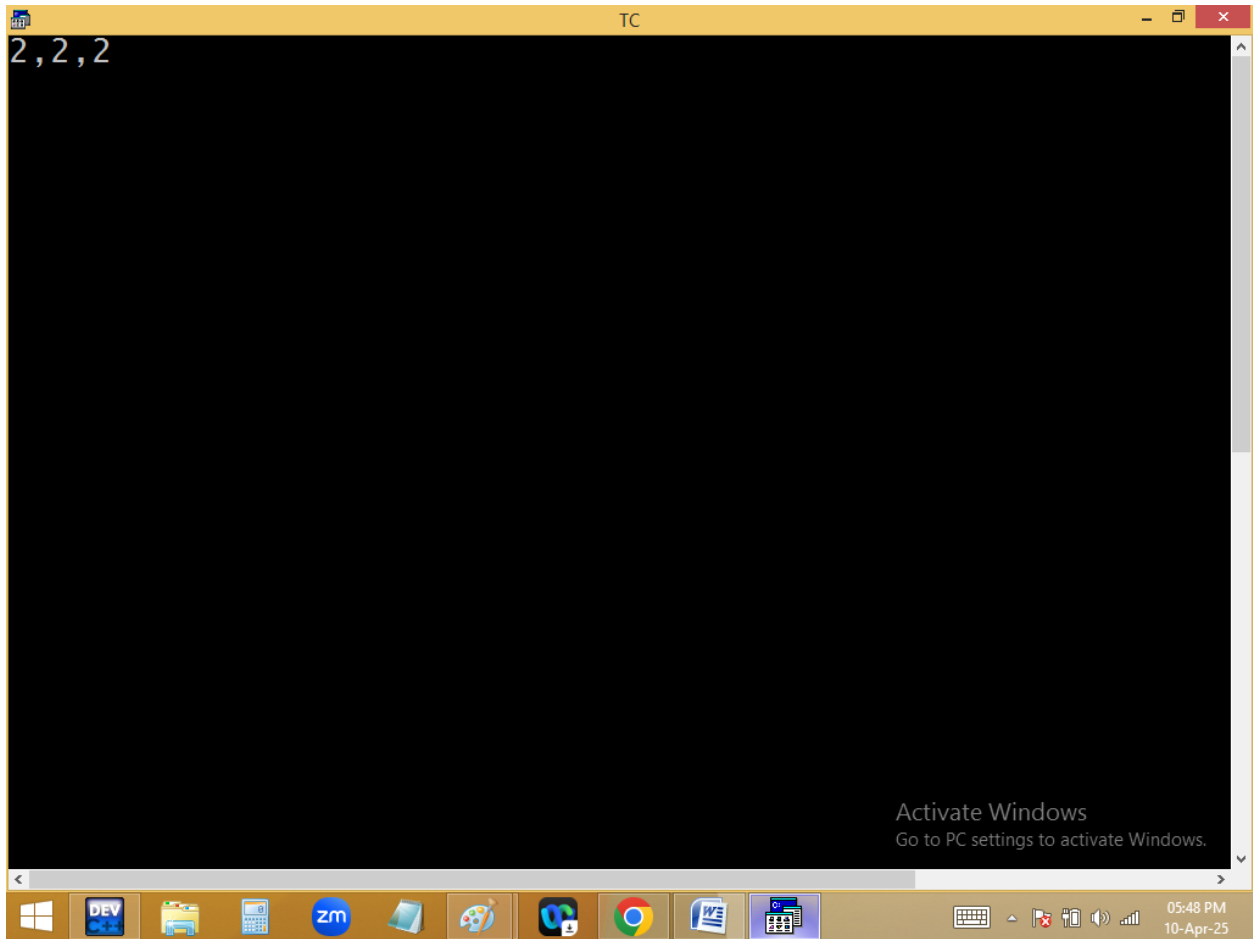


```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 49 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int *a; float *b; char *c;
clrscr();
printf("%d,%d,%d",sizeof(a),sizeof(b),sizeof(c));
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

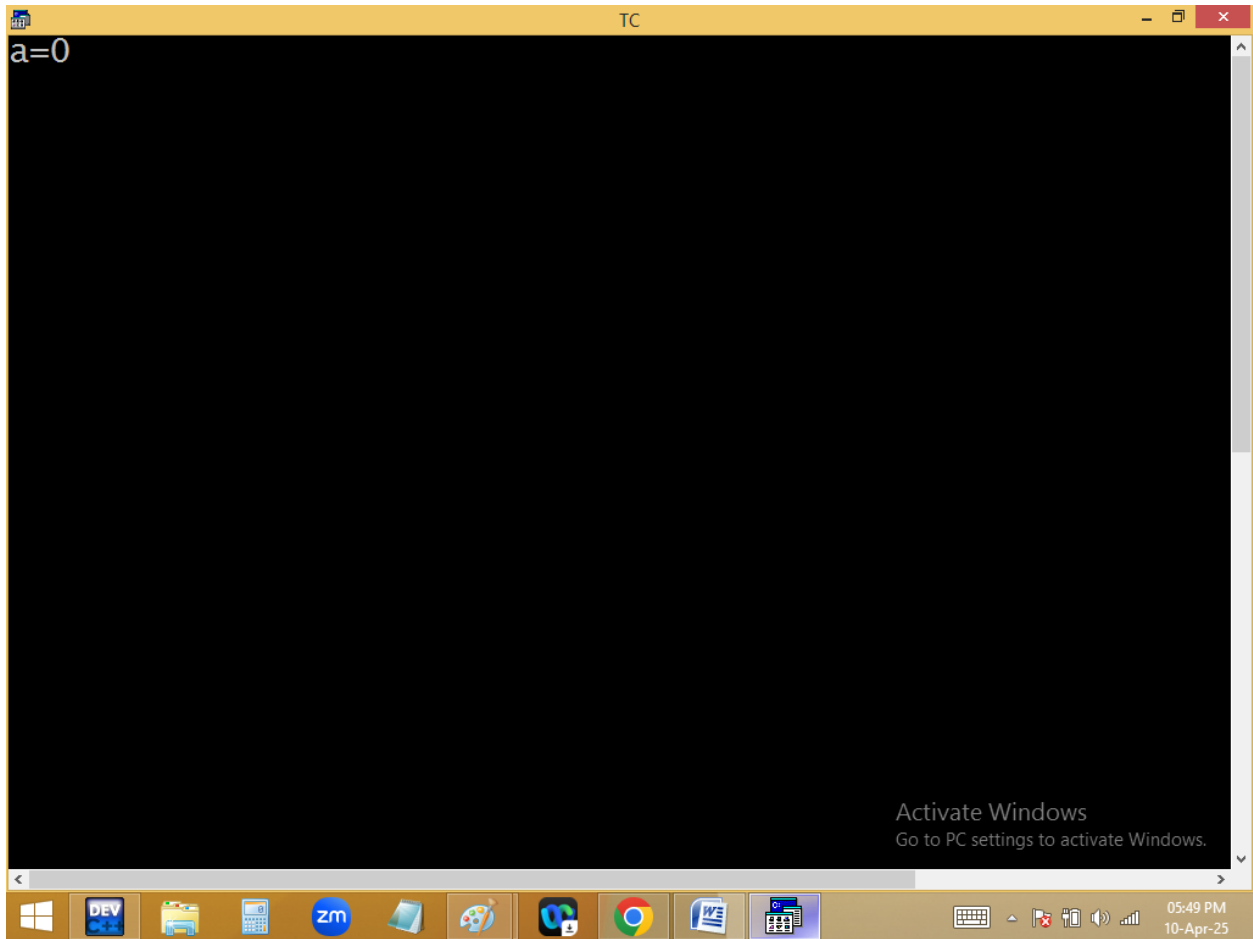
Windows taskbar: DEV, File Explorer, Calculator, ZOOM, Paint, VS Code, Chrome, Word, Excel, System tray (05:48 PM, 10-Apr-25)

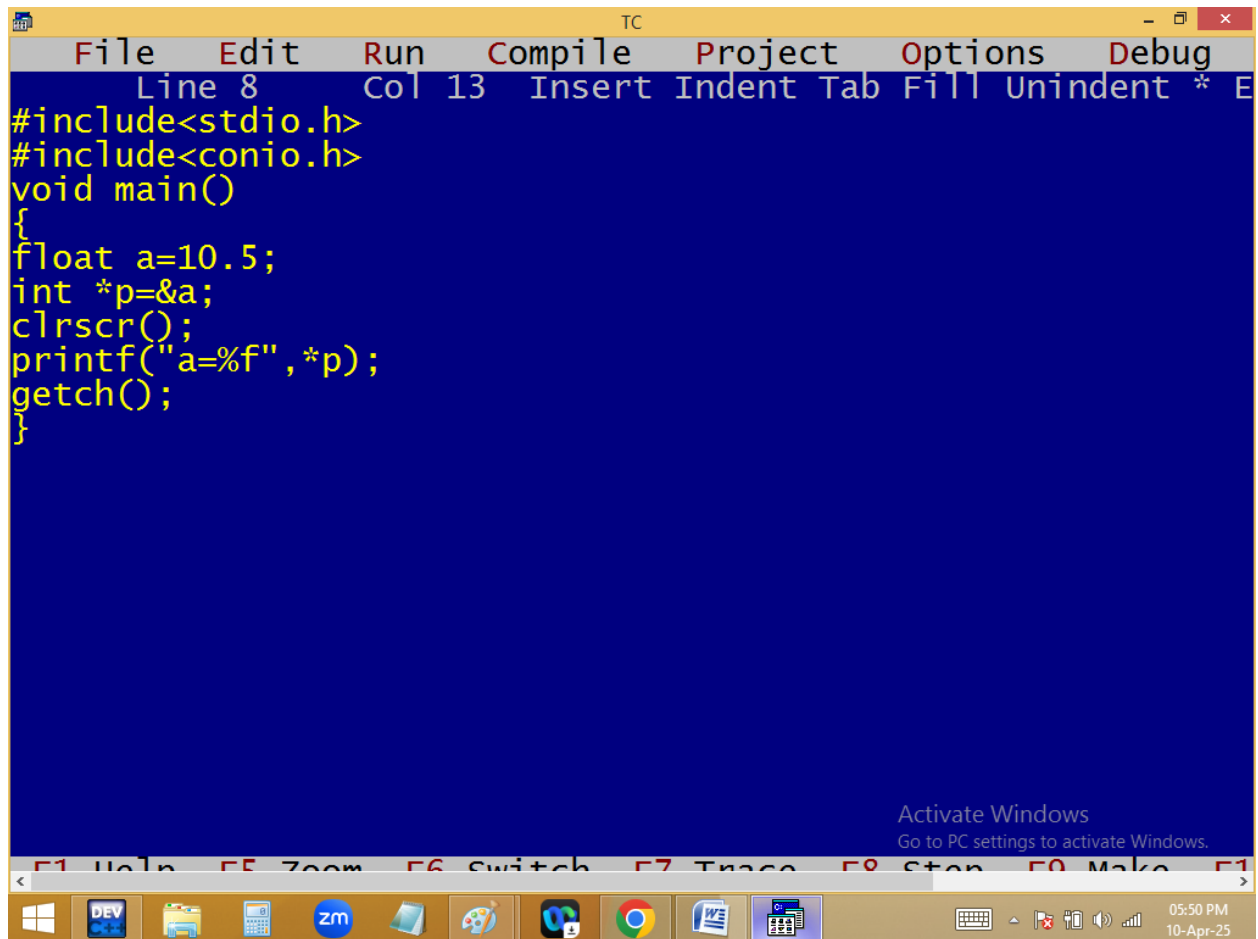


Pointer compatibility: When we are trying to store different type of address in a pointer, it gives garbage or runtime error.

```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100;
float *p=&a;
clrscr();
printf("a=%d",*p);
getch();
}

Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```





```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 13 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
float a=10.5;
int *p=&a;
clrscr();
printf("a=%f",*p);
getch();
}

Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

